

Answer Key & Teacher Notes

Session 1 – Graphing Ratios on the Coordinate Plane • Session 2 – Interpreting Points on Ratio Graphs

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Graphing Ratios on the Coordinate Plane

TALK PROMPTS

- What does the origin (0, 0) mean in the soccer-bag situation?
- Why do points made from equivalent ratios form a straight line?

PARTNER TASK

Hiking Trail, File Downloads, Street Trees. One partner writes the ordered pairs; the other plots them and checks that each point matches its table column, x first. Switch roles for each problem.

WATCH FOR

- Switching x and y, for example plotting (50, 1) for 1 hour at 50 miles per hour.
- Using uneven steps on an axis, so the points no longer line up.
- Thinking a ratio graph does not have to pass through (0, 0).

SESSION 2 • TEACHER NOTES

Interpreting Points on Ratio Graphs

TALK PROMPTS

- How does the point (1, r) show the unit rate on a ratio graph?
- How can the unit rate point help you find the pay for 11 driveways without drawing the graph?

PARTNER TASK

Water Pump, Robot Speed, Snow Shoveling. One partner reads a point aloud in words using the axis labels; the other finds the unit rate point and predicts a new point. Switch roles and check each prediction by multiplying.

WATCH FOR

- Reading a point backwards: (1, 8) as “8 items for \$1.”
- Dividing x by y ($3 \div 12$) instead of y by x to find the unit rate.
- Deciding a point is on the line without checking that $y \div x$ matches the unit rate.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.RP.A.3a	1	1	1	1	2	2	2	3	3	1	1
S1 • Form B	6.RP.A.3a	1	1	1	1	2	2	2	3	3	1	1
S2 • Form A	6.RP.A.3a	1	1	2	2	1	2	2	3	3	1	1
S2 • Form B	6.RP.A.3a	1	1	2	2	1	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

coordinate plane: A **grid** where you plot points.

ordered pair: (x, y) names one point: **across** first, then up.

origin: The point **(0, 0)** where the axes cross.

- Each column of a ratio table becomes one **ordered pair**: x from the top row, y from the bottom row.
- Points from equivalent ratios lie on a **straight** line through the **origin**.

Example: Table: 1 tray = 8 muffins, 2 trays = **16**, 3 trays = **24**. Pairs: (1, 8), (2, 16), (3, 24). Go across x, then up y. Plot each pair, starting from (0, 0). Answer: The points lie on a **straight line** through (0, 0).

YOU TRY

- (1, 4), (2, 8), (3, 12) on a line through (0, 0)
- a. (2, 14) b. (5, 35)

EXIT TICKET

- a. (2, 18) b. (4, 36)
- Yes. Each point shows 2 for every 1, so they line up with (0, 0).
- B** ((0, 0))

PRACTICE FORM A

- a. (1, 6) b. (3, 18) c. (4, 24) d. (6, 36)
- a. 6, 9, 12 b. points (1, 3), (2, 6), (3, 9), (4, 12) on a line through (0, 0) c. (6, 18)
- a. 15 b. 25 GB
- a. 14, 21 b. (6, 42)
- a. 8, 12, 20 b. $y = 32$ dollars c. points (1, 4), (2, 8), (3, 12), (5, 20), (8, 32) on a line through (0, 0)
- a. $y = 8x$ b. In 0 minutes, 0 posters are printed.
- (3, 45), because $3 \times 15 = 45$ ($2 \times 15 = 30$, $4 \times 15 = 60$)
- a. A: \$24, B: \$30 b. Store B. Its unit rate is \$2.50 per notebook, more than A's \$2.00, so its cost climbs faster.
- She switched x and y. Hours go first, so the points are (1, 50) and (2, 100).
- A** (x-value)
- B** ((3, 18))

Self-check word ORIGIN

PRACTICE FORM B

- a. (1, 8) b. (3, 24) c. (5, 40) d. (7, 56)
- a. 8, 12, 16 b. points (1, 4), (2, 8), (3, 12), (4, 16) on a line through (0, 0) c. (7, 28)
- a. 9 b. 15 GB
- a. 18, 27 b. (5, 45)
- a. 10, 20, 30 b. $y = 35$ dollars c. points (1, 5), (2, 10), (4, 20), (6, 30), (7, 35) on a line through (0, 0)
- a. $y = 6x$ b. In 0 minutes, 0 posters are printed.
- (4, 48), because $4 \times 12 = 48$ ($3 \times 12 = 36$, $5 \times 12 = 60$)
- a. A: \$25, B: \$30 b. Store B. Its unit rate is \$3.00 per notebook, more than A's \$2.50, so its cost climbs faster.
- He switched x and y. Hours go first, so the points are (1, 60) and (2, 120).
- B** (y-value)
- C** ((5, 20))

Self-check word POINTS

GUIDED NOTES

unit rate point: The point (1, r). The r is the amount for **one**.

origin: (0, 0): zero of one quantity goes with **zero** of the other.

interpret: To say in **words** what a point means.

- A point (x , y) means x of the first quantity goes with **y** of the second. Read the axis labels.
- To find the unit rate point (1, r), divide **y** by x .

Example: Read the point: 4 pounds cost **\$12**. Divide: $12 \div 4 = 3$ dollars per pound, so the unit rate point is (1, 3). Multiply: $6 \times 3 = 18$. Answer: 6 pounds cost **\$18**; the point is (6, 18).

YOU TRY

1. a. (1, 7) b. (1, 4.5)
2. In 3 hours, \$45 is earned. $45 \div 3 = \$15$ per hour.

EXIT TICKET

1. a. (1, 7) b. (1, 1.5)
2. 2 tickets cost \$10. One ticket costs $10 \div 2 = \$5$.
3. **B** (none of either quantity)

PRACTICE FORM A

1. a. (1, 5) b. (1, 9) c. (1, 2.5) d. (1, 8)
2. a. 0 bracelets need 0 charms. b. 1 bracelet needs 5 charms (the unit rate). c. $9 \times 5 = 45$ charms
3. a. 8 gallons per minute b. In 5 minutes, 40 gallons are pumped. c. $72 \div 8 = 9$ minutes
4. $11 \times 9 = \$99$
5. (8, 96): $8 \times 12 = 96$ meters
6. a. (1, 4) b. $10 \times 4 = 40$ m
7. (1, 7); yes, because $9 \times 7 = 63$
8. a. Rider 1: 12 km per hour; Rider 2: 10 km per hour
b. Rider 1; she covers more distance each hour. c. $72 - 60 = 12$ km
9. Noah read the point backwards. x is items and y is dollars, so (1, 8) means 1 item costs \$8.
10. **C** (unit rate)
11. **A** (4)

Self-check word GRAPHED

PRACTICE FORM B

1. a. (1, 7) b. (1, 6) c. (1, 1.5) d. (1, 9)
2. a. 0 bracelets need 0 charms. b. 1 bracelet needs 4 charms (the unit rate). c. $7 \times 4 = 28$ charms
3. a. 6 gallons per minute b. In 5 minutes, 30 gallons are pumped. c. $48 \div 6 = 8$ minutes
4. $7 \times 12 = \$84$
5. (6, 66): $6 \times 11 = 66$ meters
6. a. (1, 5) b. $9 \times 5 = 45$ m
7. (1, 6); yes, because $8 \times 6 = 48$
8. a. Rider 1: 15 km per hour; Rider 2: 12 km per hour
b. Rider 1; he covers more distance each hour. c. $75 - 60 = 15$ km
9. Priya read the point backwards. x is tickets and y is dollars, so (1, 6) means 1 ticket costs \$6.
10. **B** (3 boxes hold 24 pens)
11. **C** (7)

Self-check word CHARTED

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • GRAPHING RATIOS ON THE COORDINATE PLANE

1. Oranges: 3, 6, 9, 12; pairs (1, 3), (2, 6), (3, 9), (4, 12)
2. Points (1, 3), (2, 6), (3, 9), (4, 12) lie on a straight line through (0, 0). (0, 0) means 0 bags hold 0 oranges.
3. a. (7, 14) b. (9, 18)
4. a. (8, 40): $8 \times 5 = 40$ plants b. (0, 0): 0 rows have 0 plants
5. No. (1, 5) and (2, 10) show 5 per 1, but $3 \times 5 = 15$, not 14. The point (3, 14) is not on the line.
6. She switched x and y. Tickets go first: 6 tickets cost $6 \times \$2 = \12 , so the point is (6, 12).

SHEET 2 • INTERPRETING POINTS ON RATIO GRAPHS

1. a. In 2 hours the cyclist rides 24 miles. b. 12 miles per hour c. $6 \times 12 = 72$ miles
2. a. 48 b. 5 c. 84 d. 3 e. 120 f. 8
3. a. $10 \div 4 = \$2.50$ per pound b. (6, 15): $15 \div 6 = 2.5$, yes. (8, 18): $18 \div 8 = 2.25$, no (8 lb would cost \$20).
4. x is laps, so (3, 12) means 3 laps take 12 minutes (unit rate: $12 \div 3 = 4$ minutes per lap).
5. **A** ((1, 8))
6. **A** (\$9 per hour)